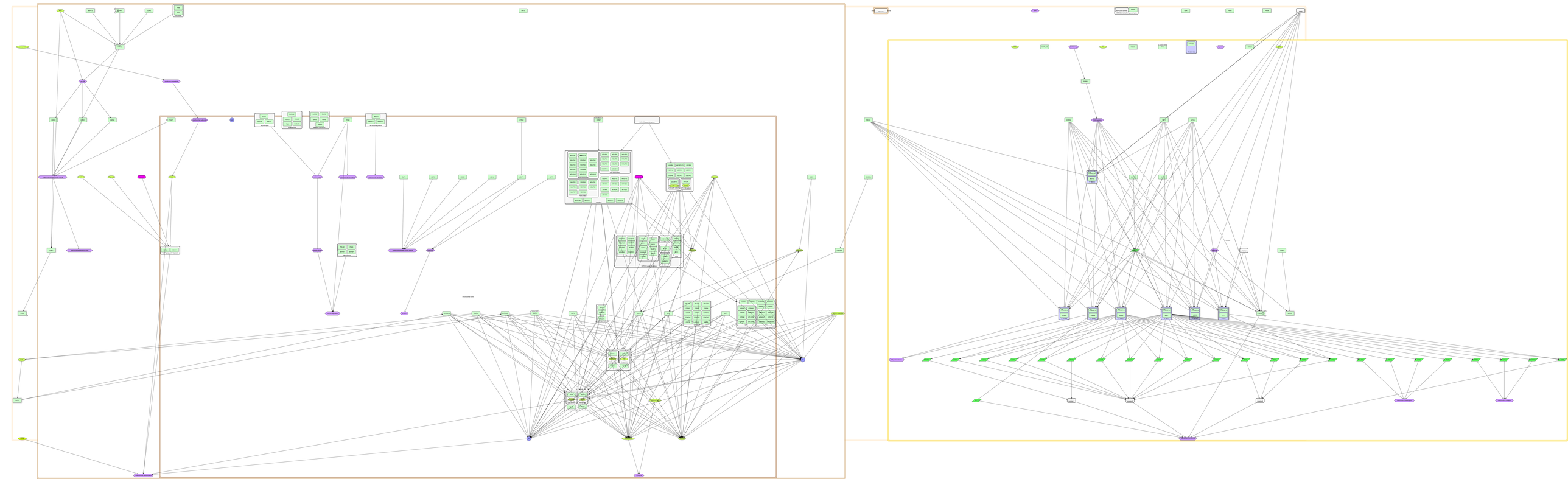

SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
UPS
NDN
PPARGC1A
PINK1
PRKN
UPS
AMPK
CAPN1
PROX1
ZNF746
ZNF746
nucleus
PARP1
DNA damage
PARP1
SIRT1
SNCA
ZNF746
GSK3B
NAD depletion
NAD(+)
PPARGC1A
PPARGC1A
PPARGC1A
S538
T177
MAPK p38
KAT2A
ESRRA
PPARA
mTORC1
MEF2D
autophagy
CDK5
mitochondrial biogenesis
T295
PPARGC1A
PPARA
TF PPARA
ESRRA
TF ERRa
YY1
TF YY1
GABPA
TF NRF2
NRF1
TF NRF1
MEF2D
TF MEF2D
MEF2C
TF MEF2C
PPARGC1A
fatty acid oxidation
SIRT3
UCP2
IDH3A
IDH3B
IDH3G
NDUFS8
ATP5MC1
COX5A
CYCS
complex IV
complex II
complex V
mitochondrial biogenesis
COX4I1
COX5A
COX5B
COX6A1
COX6B1
COX6C
COX7A2
COX7B
COX7C
COX8A
CYCS
SDHB
SDHC
SDHD
ATP5F1B
ATP5F1C
POLRMT
TFAM
TFB2M
TFB1M
mitochondrial transcription
SURF1
VDAC1
TOMM20
mitochondrial transport
RNS
NO
MEF2C
MEF2C
ROS
mutated A53T
SNCA
MEF2C
CHCHD2
TF CHCHD2
hypoxia
COX4I2
CHCHD2
ROS
CHCHD4
CHCHD2
SLC25A27
SLC25A14
SLC25A27
SLC25A14
SLC25A27
SLC25A14
ROS
PARK7
ROS
peroxynitrite
CCCP
SIRT3
NDUFA9
NDUFA10
PINK1
O2
superoxide
hydrogen peroxide
CHCHD2
H+
SOD1
mitochondrial depolarization
membrane permeability
superoxide
purine nucleotide
unsaturated fatty acid
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
mutated A53T
SNCA
UQCRH
UQCRFS1P1
UQCRQ
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron sulfur subunit
MT-CYB
heme b
MT-CYB:heme b
complex III
rotenone
superoxide
superoxide
SIRT5
CHCHD2:CHCHD10:CYCS
MT-CO1
MT-CO2
MT-CO3
COX6B1
COX7A2L
COX7A1
COX7C
COX8A
complex IV
ATP5PF
ATP5MC1
ATP5MC2
ATP5MC3
ATP5PD
ATP5MG
ATP5PB
ATP5MF
ATP5ME
MT-ATP6
MT-ATP8
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5IF1
F(0)
F(1)
complex V
tiopronin
diazoxide
ATP
ATP-sensitive K+ channel
ROS
mitochondrial dysfunction
Ca2+
UCP2
UCP3
superoxide
SDHB
SDHA
Fe-S cluster
FAD
iron sulfur protein
flavoprotein
SDHC
SDHD
complex II
SIRT3
SDHB
SDHA
Fe-S cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
ubiquinol
succinic acid
TCA cycle
ubiquinone
fumaric acid
OXPHOS assembly factors
SDHB
SDHA
Fe-S cluster
FADH2
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
CI
MT-CYB
CIII
MT-CO1
MT-CO2
MT-CO3
CIV
MT-ATP6
MT-ATP8
CV
OXPHOS subunits encoded in mtDNA
MT-TE
MT-TL1
MT-TS1
MT-TL2
MT-TS2
MT-TF
MT-TN
MT-TT
MT-TH
MT-TP
MT-TV
MT-TI
MT-TQ
MT-TW
MT-TK
MT-TR
MT-RNR1
mitochondrial tRNAs and rRNA
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
mtDNA replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
mRNA
mitochondrial DNA
mtDNA repair
mitochondrial translation
mitochondrial transcription
TFAM
POLG
ERCC6
ERCC8
Mt-DNA repair
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
nucleoid
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
mitochondrial matrix
mitochondrion
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
peroxisome
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
CDK5
mutated A141S
HTRA2
mutated G399S
HTRA2
Y182
T180
MAPK14
MAPK12
PARL
HAX1
HAX1:PARL
S400
N-3
HTRA2
HTRA2
HTRA2
HAX1
ROS
PINK1
TRAP1
PINK1
TRAP1
HTRA2
LONP1
CLPB
HSPD1
HSPE1
HSPA9
chaperone-mediated protein folding
CLPP
proteolysis
mt UPR
PINK1
TFAM
mt UPR
HSPD1
HSPE1
HSPA9
chaperone-mediated protein folding
mitochondrial respiratory chain
PINK1
PRKN
Ub
PINK1
PRKN
PINK1:PRKN
Respiratory Chain Components
ribosome
mitochondrion-attached ribosome
TOMM20
&
PINK1
ribosome
ribosome
nuclear RCC RNA
PUM1
nuclear RCC complex
nuclear RCC RNA
PINK1
nuclear RCC complex
TOMM20
PINK1
PUM1
&
nuclear RCC RNA
PUM1
nuclear RCC complex
HNRNPF
PINK1
&
PRKN
nuclear RCC RNA
PUM1
HNRNPF
nuclear RCC complex
neuron

ROS
MAPK12
MAPK14
CDK5
PARL
HAX1
HAX1:PARL
SIRT3
S400
HTRA2
peroxynitrite
mt UPR
membrane permeability
HSPE1
HSPD1
HSPA9
TRAP1
mitochondrial dysfunction
Ca2+
POLG
ERCC6
ERCC8
Mt-DNA repair
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
TFAM
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
HTRA2
PARK7
OXPHOS assembly factors
chaperone-mediated protein folding
ATP
Tiopronin
diazoxide
ROS
mtDNA repair
mitochondrial transcription
mitochondrial translation
CLPB
HSPD1
HSPE1
HSPA9
LONP1
CLPP
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
rotenone
UQCRH
UQCRFS1P1
UQCRQ
BCS1L
UQCR10
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron-sulfur subunit
MT-CYB
heme b
MT-CYB:heme b
complex III
SOD1
CHCHD4
PINK1
mitochondrial respiratory chain
KCNJ8
KCNJ11
ATP-sensitive K+ channel
mtDNA damage
POLG2
POLG
MTFMT
MTPAP
Mt replication
chaperone-mediated protein folding
proteolysis
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
superoxide
superoxide
CHCHD2
mitochondrial matrix
PRKN
mtDNA replication
mt UPR
SLC25A14
SNCA
SLC25A27
mutated A53T
SNCA
SIRT5
GHITM
CHCHD2
CYCS
CHCHD2:GHITM:CYCS
UCP3
UCP2
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX7A1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
SIRT3
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MF
ATP5PO
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
F(0)
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
complex V
purine nucleotide
SDHB
SDHA
Fe-S cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
SDHB
SDHA
Fe-S cluster
FADH2
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
succinic acid
ROS
PARK7
H+
H+
H+
fumaric acid
ubiquinol
CCCP
mitochondrial depolarization
TCA cycle
mitochondrion
ribosome
UPS
SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
NDN
PINK1
PRKN
AMPK
ROS
MAPK p38
DNA damage
NO
MEF2C
mutated A53T
SNCA
CHCHD2
TF CHCHD2
hypoxia
GSK3B
RNS
PARP1
PROX1
ESRRA
NAD depletion
SIRT1
KAT2A
CHCHD4
PPARGC1A
MEF2C
TF MEF2C
ZNF746
SNCA
PPARGC1A
PPARA
TF PPARA
nucleus
PPARGC1A
autophagy
mTORC1
CDK5
PPARGC1A
PPARA
TF PPARA
PPARGC1A
ESRRA
TF ERRa
PPARGC1A
GABPA
TF NRF2
PPARGC1A
NRF1
TF NRF1
PPARGC1A
MEF2D
TF MEF2D
PPARGC1A
YY1
TF YY1
S570
PPARGC1A
T177
MEF2D
fatty acid oxidation
ATP5F1B
COX8A1
COX7C
COX8A
COX6B1
ATP5F1C
COX5B
COX7A2
COX7B
COX4I1
COX6C
SDHD
SDHC
SDHB
COX5A
POLRMT
TFB1M
TFAM
TFB2M
SURF1
VDAC1
TOMM20
CYCS
SIRT3
complex V
complex IV
complex II
mitochondrial transcription
mitochondrial transport
mitochondrial biogenesis